

OMKARESHWARI MASHETTE

Data Scientist | Machine Learning Engineer | Generative AI & RAG Developer

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PROFESSIONAL SUMMARY

B.E. Computer Science (Data Science) graduate with hands-on experience in Machine Learning, Data Analysis, NLP, and Generative AI (GenAI) development. Skilled in Python, SQL, Scikit-learn, XGBoost, LangChain, LlamaIndex, and Retrieval-Augmented Generation (RAG). Built end-to-end AI pipelines — data ingestion, embeddings, vector search, and LLM-based QA systems — using Azure OpenAI and FastAPI. Strong foundation in statistical analysis, model evaluation, and MLOps through internship and academic projects in document retrieval, classification, and predictive modeling.

CORE COMPETENCIES

Machine Learning · Deep Learning · Data Science · Generative AI (GenAI) · Large Language Models (LLM) · Retrieval-Augmented Generation (RAG) · NLP · Prompt Engineering · Vector Databases · Predictive Modeling · Statistical Analysis · Data Preprocessing · Feature Engineering · EDA · REST API Development · Cloud Computing (Azure) · MLOps · Model Evaluation & Deployment

TECHNICAL SKILLS

Programming: Python, SQL, Git
Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Clustering, NLP, Model Evaluation, Statistical Analysis, Predictive Modeling
Generative AI: RAG, Prompt Engineering, Embeddings, Vector Search, LLM
Frameworks & APIs: LangChain, LlamaIndex, FastAPI, Flask, REST API Development
Databases & Search: MySQL, FAISS, ChromaDB, Azure AI Search
Data Analysis: Pandas, NumPy, EDA, Data Visualization, Feature Engineering, Data Preprocessing
Cloud & MLOps: Azure OpenAI, Azure ML, MLflow, GitHub, Power BI

INTERNSHIP EXPERIENCE

GenAI Intern — RAG-Based Systems | NEXACORE AI TECHNOLOGIES PVT LTD Jan 2026 – Present

- Built a document search and question-answering system using Python, LangChain, LlamaIndex, and Azure OpenAI for GenAI/LLM-driven workflows.
- Developed data ingestion, chunking, embedding, and vector indexing pipelines with Azure AI Search, FAISS, and ChromaDB for scalable RAG systems.
- Designed retrieval workflows and prompt engineering templates, improving response accuracy and document relevance.
- Built and deployed REST APIs with FastAPI; performed data preprocessing, model evaluation, and experiment tracking using Scikit-learn and MLflow (MLOps).

PROJECTS

GenAI Operations & Policy Assistant (RAG)

Python, Azure OpenAI, LangChain, Azure AI Search, FAISS

- Built a RAG-based virtual assistant for airline operations using operational manuals/SOPs; optimized accuracy via prompt engineering and metadata-based semantic search.

Adverse Event Seriousness Classification

Python, Scikit-learn, XGBoost, MLflow, SQL

- Built a supervised ML classification model to predict adverse event seriousness; engineered features from narrative text and structured data, validated with stratified techniques, and implemented automated batch scoring.

ShadowLock – AI-Enhanced Steganography Platform

Python, OpenCV, Flask, NumPy, MySQL

- Built a secure Flask web app for concealing sensitive data in digital images using image processing and encryption for secure data transmission.

EDUCATION

B.E. in Computer Science & Engineering (Data Science) — BKIT Engineering College, Bhalki | CGPA: 8.1 2022 – 2026

Pre-University Course (Science) — CB Gurukul PU College, Bhalki | 68% 2020 – 2022

CERTIFICATIONS

AI/ML with Data Science — Knowx Innovation (2026) | AI Tools & ChatGPT Workshop — be10x (2026) | Programming in Python — Echo Brains (2024)